

# SE- Introduction to Web Technologies - HTML and CSS

Session 8: Responsive Design Coverage: Media queries, Viewport, Breakpoints. Demo: Make the profile card mobile-friendly.

Task :

Question :- 1

Create a Flipkart-style product card in HTML and CSS, then use a meta viewport tag to ensure it scales correctly on mobile devices.

Ans:-

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Document</title>

  <style>

    .card{

      display: flex;

      border: 1px solid black;

      padding: 20px;

      width: 700px;

      margin: auto;

    }

    .card img{

      width: 250px;

      margin-right: 20px;
```

```
    }
  </style>
</head>
<body>
  <div class="card">
    
    <div>
      <h2>Lenovo LOQ 15IAX9</h2>
      <p>Gaming Laptop</p>
      <h3>₹89,999</h3>
    </div>
  </div>
</body>
</html>
```

Question :- 2

Add media queries to your product card so that when the screen width is 600px or less, the card layout switches from horizontal (image and details side by side) to vertical (image on top, details below).

Ans:-

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
```

```
.card {  
    display: flex;  
    border: 1px solid black;  
    padding: 20px;  
    max-width: 700px;  
    margin: auto;  
}  
  
.card img {  
    width: 250px;  
    margin-right: 20px;  
}  
  
/* Mobile View */  
@media(max-width:600px) {  
    .card {  
        flex-direction: column;  
        text-align: center;  
    }  
  
    .card img {  
        margin-right: 0;  
        margin-bottom: 15px;  
        width: 100%;  
    }  
}  
  
</style>  
</head>
```

```

<body>

  <div class="card"> 

    <div>

      <h2>Lenovo LOQ 15IAX9</h2>

      <p>Gaming Laptop</p>

      <h3>₹65,999</h3>

    </div>

  </div>

</body>

</html>

```

Question :- 3

Set up three breakpoints (600px, 900px, 1200px) for your product card and change the card's font size and padding at each breakpoint to improve readability on different devices.<br><br><em><strong>Hint:</strong> Use @media only screen and (max-width: ...) in your CSS for each breakpoint.</em>

Ans:-

```

<!DOCTYPE html>

<html>

<head>

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<style>

.card{

  display: flex;

  border: 1px solid black;

  padding: 25px;

  max-width: 800px;

```

```
    margin: auto;

    font-size: 20px;
}

/* Desktop */

@media only screen and (max-width:1200px){

    .card{

        font-size: 18px;

        padding: 20px;

    }

}

/* Tablet */

@media only screen and (max-width:900px){

    .card{

        font-size: 16px;

        padding: 15px;

    }

}

/* Mobile */

@media only screen and (max-width:600px){

    .card{

        flex-direction: column;

        font-size: 14px;

        padding: 10px;

    }

}

</style>

</head>

<body>
```

```

<div class="card">

  <div>

    <h2>Lenovo LOQ 15IAX9</h2>

    <p>Gaming Laptop</p>

    <h3>₹65,999</h3>

  </div>

</div>

</body>

</html>

```

Question :- 4

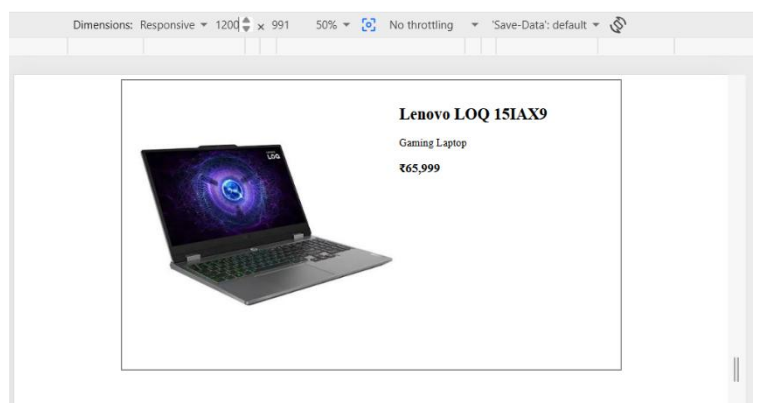
Test your responsive product card by resizing your browser window and taking screenshots at mobile, tablet, and desktop widths; submit the screenshots along with a short note on what changes at each size.

Ans:-

I tested the responsive product card by resizing the browser window and observing the layout at different screen widths.

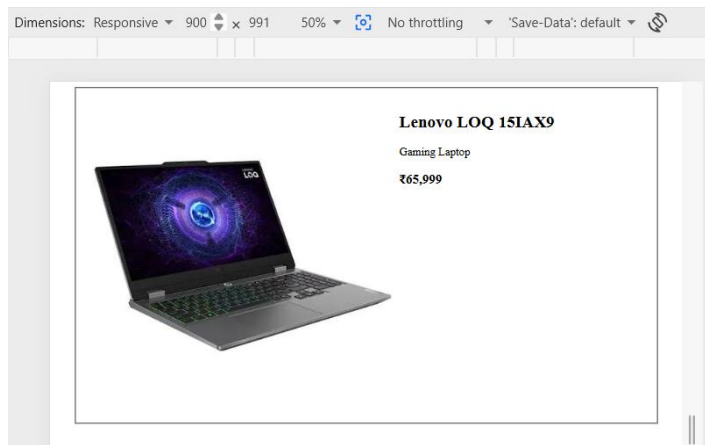
Desktop View (1200px and above):

- Layout is horizontal.
- Image and product details appear side by side.
- Larger font size and padding.



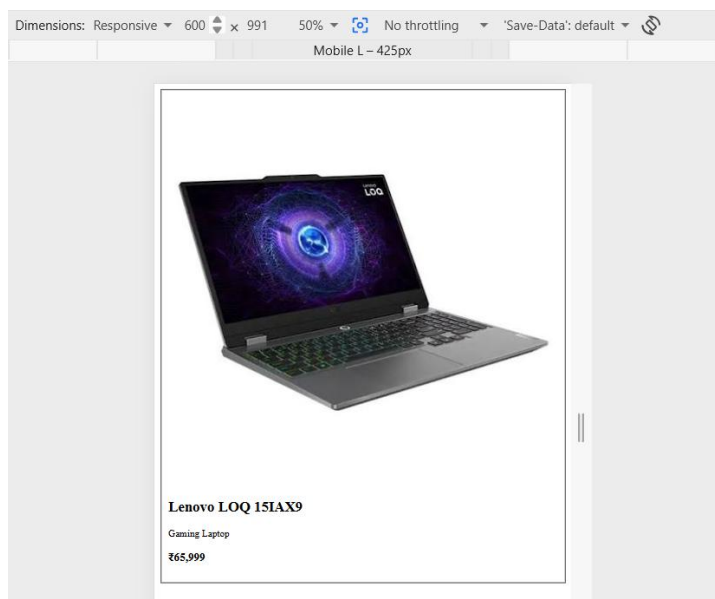
#### Tablet View (900px):

- Layout remains horizontal.
- Font size and padding are reduced slightly for better readability.



#### Mobile View (600px and below):

- Layout changes to vertical.
- Product image appears on top.
- Product details appear below the image.
- Font size and padding are reduced to fit smaller screens.



Screenshots were captured for Desktop, Tablet, and Mobile views to verify responsive behavior.

